

Acute Management of Haemophilia A and B

Definitions:

- Haemophilia A = Factor VIII deficiency
- Haemophilia B = Factor IX deficiency

Severity of Haemophilia is based on baseline Factor VIII/IX activity level:

- Severe < 1%
- Moderate 1-5%
- Mild 5-40%

Clinical Presentations of Bleeds

Bleeds can occur spontaneously in patients with severe Haemophilia i.e. without apparent preceding trauma. Hence most patients with severe Haemophilia are managed with prophylactic home factor replacement therapy to prevent bleeds.

- (a) Joint bleed/haemarthrosis
 - Hallmark of Haemophilia
 - Tingling sensation, pain, warmth, swelling of affected joint, refusal to move
 - Usually affects knees, ankles, elbows, less commonly shoulders, hips
- (b) Muscle bleed/haematoma
 - Pain, warmth, swelling of affected muscle
 - Note: Psoas haematoma may present with thigh/hip/groin pain, lower back pain, or right iliac fossa pain mimicking acute appendicitis
- (c) Intracranial haemorrhage
 - Headache, nausea, vomiting, drowsiness, seizure, focal weakness/numbness
 - Potentially life-threatening
- (d) Mucosal bleed
 - Gum/nose bleed
- (e) Gastrointestinal bleed
 - Less common
 - Haematemesis, blood in stools/melena
- (f) Haematuria (less common)

Management of Acute Bleed

- Treatment should be given *as soon as possible* after the onset of a bleed.
- ***When in doubt, treat first.*** Pain is often relieved within minutes.
- Examine area of suspected bleed.
- Tingling or “bubbling” sensation is an early symptom of joint bleed. Pain and swelling are later manifestations. Treat joint bleeds early to prevent joint damage.
- *Not necessary to do X-ray* in acute joint/muscle bleed unless fracture is suspected due to mechanism of injury.
- For head injury with no clinical evidence of intracranial bleed, administer Factor as a prophylaxis as intracranial bleed can be delayed – desired Factor level: at least 50%.
- Early haemarthroses and muscle haematomas can be treated in the outpatient setting.
- More severe bleeds e.g. intracranial bleed, psoas haematoma require admission.

Management of Acute Bleed

- Set IV cannula or use butterfly needle to administer Factor concentrate (preferably use the patient's usual brand of Factor concentrate) according to desired peak activity. Preferred site of cannulation: dorsum of hands
- Watch for acute reactions – e.g. allergy or anaphylaxis

Factor Replacement

Factor VIII Dosing

For Haemophilia A patients without inhibitors

- Advate/Xyntha (Recombinant) or Haemoctin (plasma-derived) – standard half-life products
- 1 IU/kg of Factor VIII increases Factor VIII activity by 2%
- Dose required = weight x 0.5 x desired increase in Factor level
- E.g. For severe Haemophilia A, 20 kg patient, to achieve 40%, give $20 \times 0.5 \times 40 = 400$ IU (1 vial = 500 IU)
Always round up to nearest vial to avoid wastage. Hence 1 vial of 500 IU Factor VIII is required.
- Standard half-life Factor VIII has shorter half-life (8-12 h) – frequency 8-12 hourly.
- Very expensive, do not waste.
- Read drug insert carefully and follow instructions regarding reconstitution and administration as every brand is different.
- If parents are experienced, allow parents to reconstitute and administer.

Factor IX Dosing

For Haemophilia B patients without inhibitors

- Benefix (Recombinant) or Alphanine (plasma-derived) – standard half-life products
- For Benefix, 1 IU/kg increases Factor IX activity by 0.7% in children < 12 years old and 1 IU/kg increases Factor IX activity by 0.8% in children 12 years old or older.
- For Alphanine, 1 IU/kg increases Factor IX activity by 1%
- Standard half-life Factor IX has half-life of about 18 h – frequency 24 hourly
- Very expensive, do not waste.
- Read drug insert carefully and follow instructions regarding reconstitution and administration as every brand is different.
- If parents are experienced, allow parents to reconstitute and administer.

Pain Management

- Initial first aid: RICE (rest, ice pack – 5 min on, 10 min off, compression with bandage, elevation)
- Factor replacement should relieve pain
- Paracetamol
- Oral opioid e.g. Tramadol
- COX-2 inhibitors e.g. Etoricoxib (Arcoxia)
- Avoid NSAIDS

Other adjunctive therapy

- Oral tranexamic acid (antifibrinolytic agent) 10-25 mg/kg/dose tds for mucosal bleed (gum/nose bleed)
- For dental procedures/gum bleed, dissolve 500 mg tablet in 10 ml of water and use as gargle tds or soak gauze in tranexamic solution and apply soaked gauze over bleeding gum

Recommended Dosing of Factor Replacement

Type of bleed	Haemophilia A Desired level	Hpl A FVIII dose (IU/kg)	Haemophilia B Desired level	Hpl B FIX (Benefix)* dose (IU/kg)	Duration (Days)
Joint	40-60%	20-30	40-60%	50-80	1-2
Muscle (except iliopsoas)	40-60%	20-30	40-60%	50-80	2-3
Iliopsoas					
- Initial	80-100%	40-50	60-80%	80-100	1-2
- Maintenance	30-60%	15-30	30-60%	40-80	3-5
Intracranial					
- Initial	80-100%	40-50	60-80%	80-100	1-7
- Maintenance	50%	25	30%	40	8-14
Throat and Neck					
- Initial	80-100%	40-50	60-80%	80-100	1-7
- Maintenance	50%	25	30%	40	8-14
Gastrointestinal					
- Initial	80-100%	40-50	60-80%	80-100	7-14
- Maintenance	50%	25	30%	40	
Renal	50%	25	40%	50	3-5
Deep laceration	50%	25	50%	65	5-7
Surgery (major)					
- Pre-op	80-100%	40-50	60-80%	80-100	
- Post-op	60-80%	30-40	40-60%	50-80	1-3
	40-60%	20-30	30-50%	40-65	4-6
	30-50%	15-25	20-40%	26-50	7-14
Surgery (minor)					
- Pre-op	50-80%	25-40	50-80%	65-100	
- Post-op	30-80%	15-40	30-80%	40-100	1-5

*Benefix dose in table is based on calculation for children 12 years and older,
Dose of Factor IX (Benefix) required (IU) = weight (kg) x desired Factor IX increase x 1.3
For Haemophilia B patients less than 12 years old,
Dose of Factor IX (Benefix) required (IU) = weight (kg) x desired Factor IX increase x 1.4

Management of Bleed in Haemophilia A with inhibitor

If patient with inhibitor is on Hemlibra (Emicizumab) prophylaxis, treatment of bleed can only be recombinant activated Factor VII (Novoseven) 90 mcg/kg 3-6 hourly (round up to the nearest vial).

If patient with inhibitor is not on Hemlibra prophylaxis, FEIBA 50-100 IU/kg 12 hourly can be given for treatment of bleed.

Reference:

WFH (World Federation of Hemophilia) Guidelines for the management of Hemophilia, 3rd edition (2020)